HW 5.4.2: The Derivative and Function Information

Use your knowledge of derivatives to answer the questions. A calculator should not be used.

1. Given $f(x) = x^3 - 6x^2 + 9x + 1$ answer the following:

   a. Is the function increasing or decreasing at $x = 2$?

   b. List the interval(s) on which $f(x)$ is increasing.

   c. List the interval(s) on which $f(x)$ is decreasing.

   d. At what value(s) does $f(x)$ have a relative maximum or minimum?

   e. (Optional) Sketch what you think the graph of $f(x)$ should look like based on your answers from b-d.

2. Given $f(x) = 3x^2 - 12x + 7$ answer the following:

   a. Is the function increasing or decreasing at $x = 3$?

   b. List the interval(s) on which $f(x)$ is increasing.

   c. List the interval(s) on which $f(x)$ is decreasing.

   d. At what value(s) does $f(x)$ have a relative maximum or minimum?

   e. (Optional) Sketch what you think the graph of $f(x)$ should look like based on your answers from b-d.

3. Given $f(x) = -x^3 + 3x^2 + 24x - 5$ answer the following:

   a. Is the function increasing or decreasing at $x = 3$?

   b. List the interval(s) on which $f(x)$ is increasing.

   c. List the interval(s) on which $f(x)$ is decreasing.

   d. At what value(s) does $f(x)$ have a relative maximum or minimum?

   e. (Optional) Sketch what you think the graph of $f(x)$ should look like based on your answers from b-d.

4. Given $f(x)=-2x^3+15x^2-6$ answer the following:

    a. Is the function increasing or decreasing at $x=-1$?

    b. List the interval(s) on which $f(x)$ is increasing.

    c. List the interval(s) on which $f(x)$ is decreasing.

    d. At what value(s) does $f(x)$ have a relative maximum or minimum?

    e. (Optional) Sketch what you think the graph of $f(x)$ should look like based on your answers from b-d.

5. Given $f(x)=x^4-8x^2+3$ answer the following:

    a. Is the function increasing or decreasing at $x=-1$?

    b. List the interval(s) on which $f(x)$ is increasing.

    c. List the interval(s) on which $f(x)$ is decreasing.

    d. At what value(s) does $f(x)$ have a relative maximum or minimum?

    e. (Optional) Sketch what you think the graph of $f(x)$ should look like based on your answers from b-d.

Selected Answers:

1. $f'(x) = 3x^2 - 12x + 9$

a. decreasing
b. $(-\infty,1)$ and $(3,\infty)$
c. $(1,3)$
d. $f(x)$ has a relative minimum at $x = 3$. $f(x)$ has a relative maximum at $x = 1$.

3. $f'(x) = -3x^2 + 6x + 24$

a. increasing
b. $(-2,4)$
c. $(-\infty,-2)$ and $(4,\infty)$
d. $f(x)$ has a relative minimum at $x = -2$. $f(x)$ has a relative maximum at $x = 4$.

5. $f'(x) = 4x^3 - 16x$

a. increasing
b. $(-2,0)$ and $(2,\infty)$
c. $(-\infty,-2)$ and $(0,2)$
d. $f(x)$ has a relative minimum at $x = -2$ and $x = 2$. $f(x)$ has a relative maximum at $x = 0$.